STRhub Verified

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Technical Reproducibility Validation Report

mahi-mahi

Version v1.0.0

ont-bam-hg38-med

Verification date	2026-11-06
Repository commit	861ea09ab44e88424c45b04d8018f3948a8fefbb
Result	**Runs as documented**
Reached	Runs + Plausible output
Permalink	https://strhub.app/verified/mahi-mahi-ont-med
STRhub	https://strhub.app

SCOPE OF THIS REPORT

Independent automated verification of reproducible execution. It confirms the tool installs, runs end-to-end, and produces structurally valid output in a standardized environment.

This is **not** an analytical validation. Genotype accuracy, concordance, forensic suitability, and regulatory compliance are out of scope.

WHAT WAS VERIFIED

– Source available
– Installation successful
– End-to-end execution
– Output generated

NOT VERIFIED

– Genotype accuracy
– Concordance
– Forensic validity
– Regulatory compliance

1. Executive Summary

PURPOSE	Verify that the tool installs, runs end-to-end, and produces structurally valid output.
RESULT	**Runs as documented**
REACHED	Runs + Plausible output, 6/6 gates passed
WHY	The tool installed and its run produced its documented output.
SCOPE	Installation, execution, and output structure verification.
NOT EVALUATED	Genotype accuracy · Concordance · Forensic validity · Regulatory compliance

2. Step 1: The Documented Run

As documented, it runs.

A third party installed the tool with the steps this commit states and ran the command it writes, and the output file that command names appeared. This says nothing about genotypes or about the forensic panel.

3. As it is in the repository: Verification Gates

Available	The pinned public source exists	PASS
Installs	The environment builds from source	PASS
Starts	The installed program answers --help in the built environment	PASS
Runs	Executes end-to-end without crashing	PASS
Expected IO	Produces a non-empty file in the declared format	PASS
Plausible Output	Output structure matches expected genotype-bearing format	PASS
Execution log	mahi-mahi-ont-med.log-external.txt	

4. What the Author Documents as a Known Issue

Documented by the author in the official repository. Quoted from the repository's README at the verified commit. STRhub did not establish any of this by running the tool; it is the author's own note about their own software.

Warnings, Precautions, and Limitations (Official README line 813)

Warnings and precautions - Use the loci file version that matches the reference genome passed to --ref. - Haplotagged input is recommended. If --bypass-haplotag-check is used, close-in-length alleles may be merged and accuracy may be reduced. - If using non-standard --loci, check the BED file carefully before use. - If a TR region contains sequence that is not actually repetitive, motif decomposition of this unexpected sequence may be effectively nonsensical. The reported length in bases is still meaningful. - Previous versions of mahi-mahi have used different parameters, configurations, an…

Warnings and precautions (Official README line 815)

- Use the loci file version that matches the reference genome passed to --ref. - Haplotagged input is recommended. If --bypass-haplotag-check is used, close-in-length alleles may be merged and accuracy may be reduced. - If using non-standard --loci, check the BED file carefully before use. - If a TR region contains sequence that is not actually repetitive, motif decomposition of this unexpected sequence may be effectively nonsensical. The reported length in bases is still meaningful. - Previous versions of mahi-mahi have used different parameters, configurations, and filenames. Please refer to…

Known limitations (Official README line 827)

- When the original BAM contains many non-spanning reads, `DP` can be lower than an allele's `AD` or the sum of `AD`. `DP` measures mean alignment depth over the reference repeat interval, whereas `AD` counts reads selected for allele-length estimation, including selected non-spanning reads. This difference does not by itself indicate an invalid call. - Expansions supported mainly by non-spanning reads may be underestimated and sized less precisely than fully spanned alleles. - Apparent interruptions may be sequencing errors. Visual verification using the read visualisation is recommended. - I…

Troubleshooting and Support (Official README line 891)

For help, contact Oxford Nanopore Technologies through your usual support channel: [Oxford Nanopore customer support](https://nanoporetech.com/support/customer-support). ## Common checks - Confirm that the reference genome matches the loci file. - Confirm that the BAM or CRAM file is readable and indexed. - Confirm that the sample sex argument is correct. - Confirm that the input BAM is haplotagged unless --bypass-haplotag-check is intentionally being used. ## Common problems and messages | Message or issue | Likely cause | Suggested action | |----------|----------|…

5. Exact Run Command

This is the command that was executed, verbatim, in the CI environment.

```
# Environment: ubuntu-22.04 · mahi-mahi v1.0.0 · commit 861ea09
$ cd '/opt/tool' &&
touch /tmp/.strhub_mark &&
( mkdir \
-p out &&
mahi-mahi \
--bam /data/in/input.bam \
--ref /data/ref/hg38.fa \
--loci hg38 \
--sample SAMPLE \
--sex XX \
--out-dir out \
); rc=$?; find . \
-xdev \
-type f \
-newer /tmp/.strhub_mark \
-size +0 \
! \
-path './.git/*' \
! \
-path './proc/*' \
! \
-path './sys/*' \
! \
-path './dev/*' \
! \
-path './data/*' \
2>/dev/null |
head \
-500 |
while IFS= read \
-r f; \
do mkdir \
-p "/data/out/$(dirname \
"$f")" &&
cp "$f" "/data/out/$f"; done; exit $rc
```

6. Reproducibility Metadata

Tool	mahi-mahi
Version	v1.0.0
Repository	https://github.com/nanoporetech/mahi-mahi
Commit (pinned)	861ea09ab44e88424c45b04d8018f3948a8fefbb
Container OS	ubuntu-22.04
Marker panel	ont-bam-hg38-med
Verified on	2026-11-06 16:25:27 UTC
CI run	https://github.com/Tfronta/strhub-verified/actions/runs/42251428166

Verified automatically, in a clean environment, on the tool's public source at the pinned commit. This certificate records what happened; it is not an endorsement by the tool's author.

7. Out of Scope

This report does not evaluate any of the following:

– Genotype correctness or accuracy
– Concordance against known truth sets
– Sensitivity, specificity, or stutter performance
– Allele calling accuracy or forensic casework suitability
– Regulatory compliance or ISO accreditation
– Multi-laboratory or multi-dataset reproducibility

8. Output Content Evidence

Output file	out/SAMPLE.vcf
Format	VCF
Sequence records	99
Distinct STR loci	99
Total reads	2,533
Max depth (single locus)	52

9. Verification Data

The repository ships example data (2 file(s)); this run did not use it. STRhub ran the verification using the public reference dataset listed below. A small test file in the repository lets a new user run the tool on their first day and see it working before trusting it with their own data, and it lets a verification run against the author's own sample as well as this one. Publishing the output that file should produce helps just as much: it shows what the results are meant to look like, which is what a reader needs to tell a correct run from one that merely finished.

Dataset	1000 Genomes ONT, hg38 medical STR slice (R10 SUP)
Source	https://s3.amazonaws.com/1000g-ont/index.html?prefix=PROCESS…
License	Open access (1000 Genomes / HPRC). Research use. STRhub is not the data provider.
Ref. genome	GRCh38 / hg38

Loci tested	ABCD3 · AFF2 · AFF3 · AR · ARX_EIEE · ARX_PRTS · ATN1 · ATXN1 · ATXN10 · ATXN2 · ATXN3 · ATXN7 · ATXN7_CCG · ATXN8OS · ATXN8OS_CTA · BCLAF3 · BEAN1 · C9ORF72 · CACNA1A · CBL · CNBP · CNBP_CA · CNBP_CAGA · COMP · CSTB · DAB1 · DAB1_ATTTT1 · DAB1_ATTTT2 · DIP2B · DMPK · EIF4A3 · EIF4A3_CACA20_3P · EIF4A3_CACA20_5P · EP400 · FAM193B · FGF14 · FMR1 · FOXL2 · FXN · FXN_AAAA · GIPC1 · GLS · GOLGA8A · HOXA13-I · HOXA13-II · HOXA13-III · HOXD13 · HTT · HTT_CCG · HTT_nonrepeat · JPH3 · LRP12 · MARCHF6 · MARCHF6_TTTTA · MARCHF6_TTTTA2 · MIR7-2 · NAXE · NIPA1 · NOP56 · NOP56_CGCCTG · NUTM2B-AS1 · PABPN1 · PHOX2B · PPP2R2B · PRDM12 · PRNP · PRNP_CCTCAGGGCGGTGGTGGCTGGGGGCAG · RAI1 · RAI1_TTTTA · RAI1_TTTTA2 · RAPGEF2 · RAPGEF2_TTTTA_3P · RAPGEF2_TTTTA_5P · RFC1 · RILPL1 · RUNX2 · SAMD12 · SAMD12_TTTTA_3P · SAMD12_TTTTA_5P · SOX3 · STARD7 · STARD7_AAAAT · STARD7_AAAAT2 · TBP · TBX1 · TCF4 · THAP11 · TNRC6A · TNRC6A_TTTTA_3P · TNRC6A_TTTTA_5P · TYMS · VWA1 · YEATS2 · YEATS2_TTTTA · YEATS2_TTTTA2 · ZFHX3 · ZIC2 · ZIC3 · ZNF713

10. Verification Matrix

N/A	Run on repository test data	—
	Run on reference dataset	1000 Genomes ONT, hg38 medical STR slice (R10 SUP)
	README check (5/5)	Advisory: all items present
	Install / environment setup	Present
	Run command	Present
	Expected input format	Present
	Produced output format	Present
	Dependencies / versions	Present

11. What This Run Needed Beyond the Repository

The result above describes a run configured as follows. Anyone repeating it needs the same things.

– Test data: no sample from the repository was used, so a public reference sample stood in.

– A container environment, built from the tool's declared install steps rather than from a recipe the repository ships.

12. Notes from Reading the Repository

Recorded automatically from the tool's public files when this run was configured. Not verified by execution, and not part of the gates above. Useful for what to check by hand.

– Environment: generated by STRhub from the repository's pyproject.toml (pip), at the pinned commit.
– Input: the README documents BAM (line 55); this run used STRhub's ont-bam-hg38-med reference data.
– Panel: the loci file the documented command reads, shipped in the repository, lists coordinates inside STRhub's medical STR slice and none inside the forensic one; this run used the medical slice of the same sample (ont-bam-hg38-med).
– The output directory the command writes into (out) was created first.
– 1 input path(s) in the README command replaced with /data/in/input.bam.
– 1 reference FASTA path(s) replaced with /data/ref/hg38.fa.
– Run command: the README's own command, rewritten to STRhub's mounts; everything it created was captured as output.

13. Limitations

– Single reference dataset per input type
– Single containerized environment (Docker / ubuntu-22.04)
– No truth-set comparison or ground-truth genotypes
– No accuracy or concordance assessment
– No forensic validation of results
– Short-read limitations apply (very long STR alleles may not span reads)

14. Scope and Disclaimers

Executed end-to-end in the stated environment with output in the expected format. Concerns reproducible execution only; no claim of accuracy, casework fitness, or regulatory validation.

Each result is a dated snapshot, verified on the tool's public repository at a pinned commit. STRhub does not store tool source code.

15. Conclusion

◆ **Runs end-to-end**

mahi-mahi v1.0.0 installs and executes without error in a clean ubuntu-22.04 environment at the pinned commit.

◆ **Produces plausible output**

The tool generated a structurally valid VCF output file with genotype calls across 99 target forensic STR loci, with a maximum read depth of 52 at NUTM2B-AS1.

◆ **Verdict: Runs**

The tool installed and its run produced its documented output.

◆ **Repository test data not used**

The repository ships example data (2 file(s)), which this run did not use: the gates above ran on STRhub's reference dataset. STRhub Verified supplied 1000 Genomes ONT, hg38 medical STR slice (R10 SUP) as the reference input for this verification run. Users replicating this result should use the same or equivalent publicly available reference material.

◆ **Reproducibility statement**

The exact command reported in Section 3, executed at the pinned commit in the described environment, is sufficient to reproduce this verification result independently.

Appendix: Detected Markers with Read Depth (1/3)

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LOCUS	READ DEPTH
NUTM2B-AS1	52
GLS	49
TNRC6A_TTTTA_5P	48
TNRC6A	48
TNRC6A_TTTTA_3P	48
ATXN1	47
RFC1	43
ZIC2	43
LRP12	42
ATXN2	42
PABPN1	42
HTT	41
HTT_nonrepeat	41
HTT_CCG	41
CBL	40
AFF3	39
SAMD12_TTTTA_3P	39
SAMD12	39
SAMD12_TTTTA_5P	39
ATXN8OS_CTA	39
ATXN8OS	39
NOP56	39
NOP56_CGCCTG	39
DAB1_ATTTT1	38
DAB1	38
DAB1_ATTTT2	38
RAPGEF2_TTTTA_5P	38
RAPGEF2	38
RAPGEF2_TTTTA_3P	38
PPP2R2B	37
RUNX2	37
CACNA1A	37
ATXN10	37
MARCHF6	36
MARCHF6_TTTTA2	36
FXN_AAAA	36
FXN	36
DIP2B	36
GIPC1	36
MARCHF6_TTTTA	35

Appendix: Detected Markers with Read Depth (2/3)

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LOCUS	READ DEPTH		LOCUS	READ DEPTH	
BEAN1	35	■■■■■■■■■■	STARD7_AAAAT2	25	■■■■■■■
C9ORF72	34	■■■■■■■■■■	STARD7	25	■■■■■■■
TCF4	34	■■■■■■■■■■	STARD7_AAAAT	25	■■■■■■■
DMPK	34	■■■■■■■■■■	ARX_PRTS	25	■■■■■■■
CSTB	34	■■■■■■■■■■	ARX_EIEE	25	■■■■■■■
YEATS2_TTTTA	33	■■■■■■■■■■	CNBP	24	■■■■■■■
ATXN3	33	■■■■■■■■■■	CNBP_CAGA	24	■■■■■■■
JPH3	33	■■■■■■■■■■	CNBP_CA	24	■■■■■■■
COMP	33	■■■■■■■■■■	FOXL2	23	■■■■■■■
YEATS2	32	■■■■■■■■■	PHOX2B	20	■■■■■■
YEATS2_TTTTA2	32	■■■■■■■■■	AR	20	■■■■■■
ATN1	31	■■■■■■■■■	FMR1	16	■■■■
HOXD13	29	■■■■■■■■	AFF2	15	■■■■
TBP	29	■■■■■■■■	SOX3	13	■■■■
HOXA13-III	29	■■■■■■■■	VWA1	0	
HOXA13-II	29	■■■■■■■■	ABCD3	0	
NIPA1	29	■■■■■■■■	NAXE	0	
HOXA13-I	28	■■■■■■■■	FAM193B	0	
ATXN7	26	■■■■■■■■	ZNF713	0	
ATXN7_CCG	26	■■■■■■■■	PRDM12	0	

Appendix: Detected Markers with Read Depth (3/3)

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LOCUS	READ DEPTH
RILPL1	0
EP400	0
FGF14	0
GOLGA8A	0
MIR7-2	0
THAP11	0
ZFHX3	0
RAI1_TTTTA	0
RAI1	0
RAI1_TTTTA2	0

LOCUS	READ DEPTH
EIF4A3_CACA20_3P	0
EIF4A3	0
EIF4A3_CACA20_5P	0
TYMS	0
PRNP_CCTCAGGGCGGTGGTGGCTGGGGGCAG	0
PRNP	0
TBX1	0
BCLAF3	0
ZIC3	0